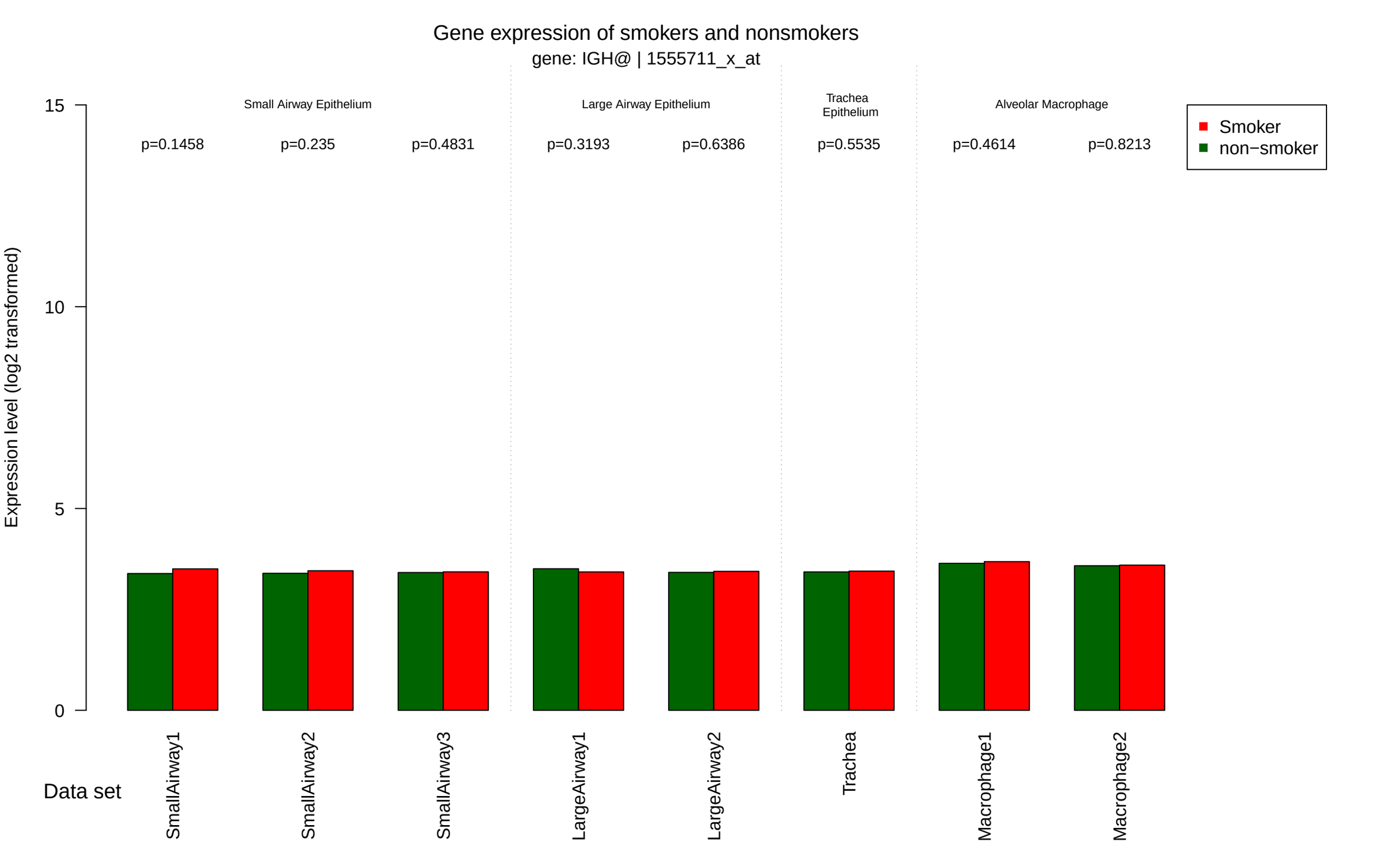
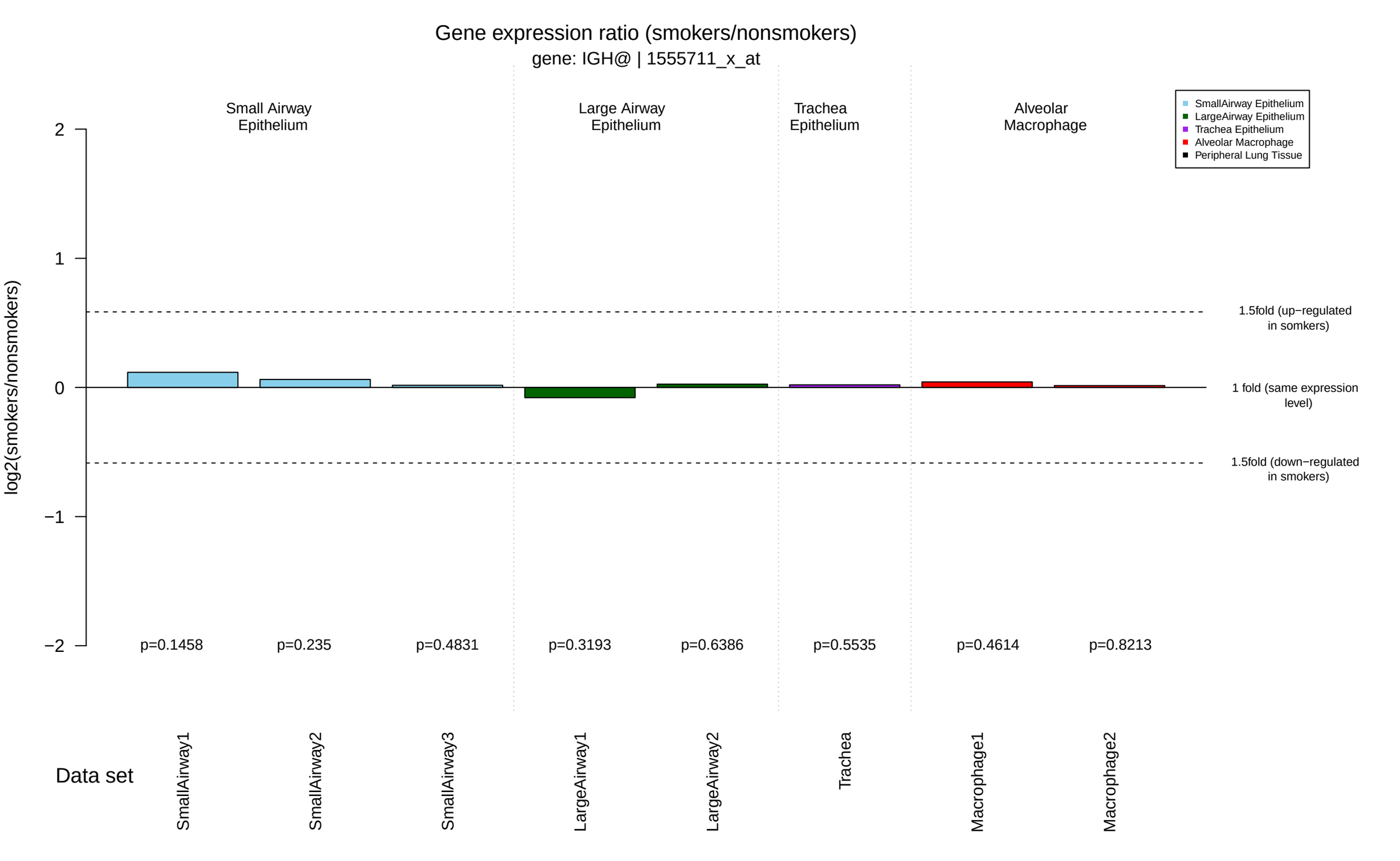
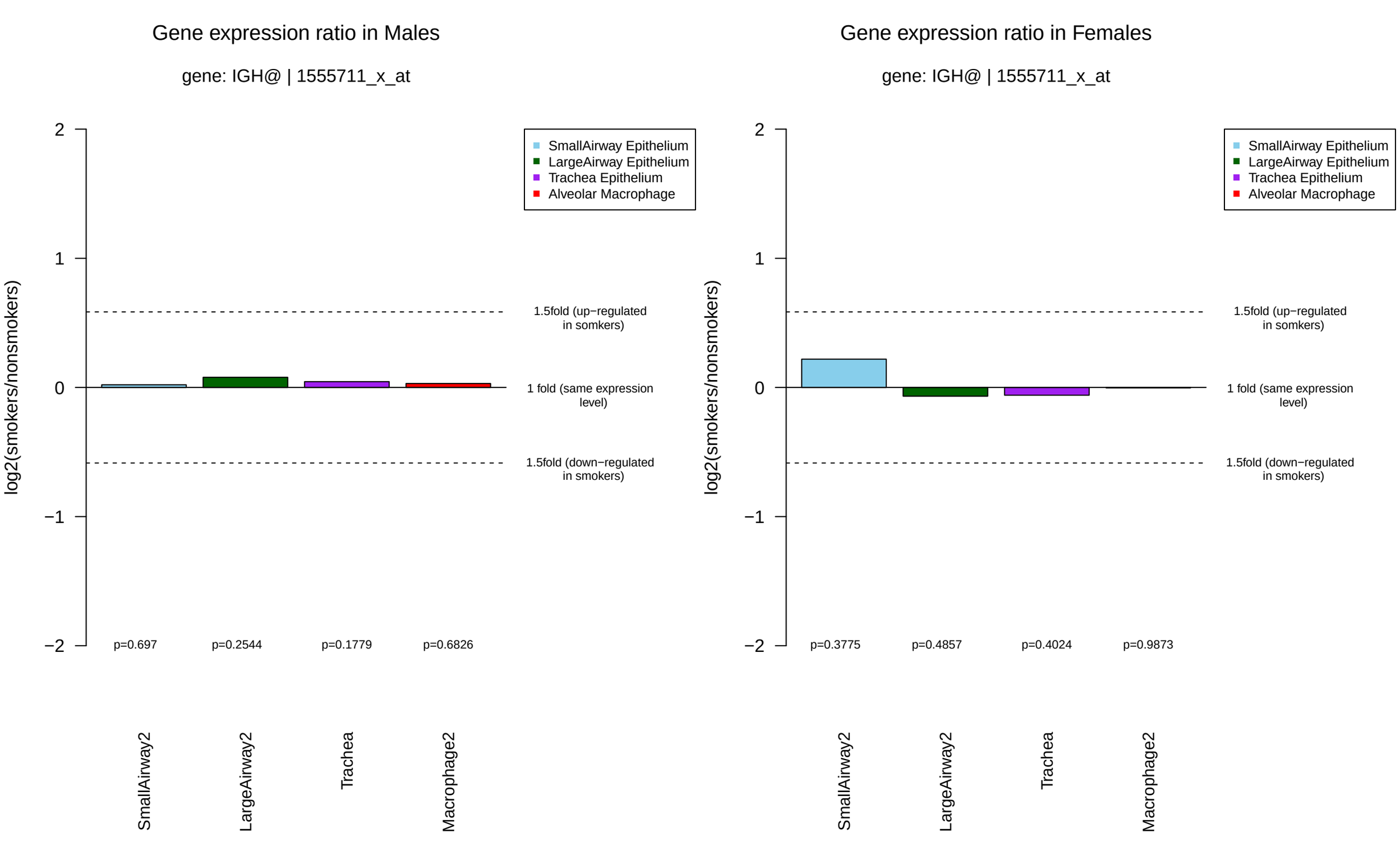
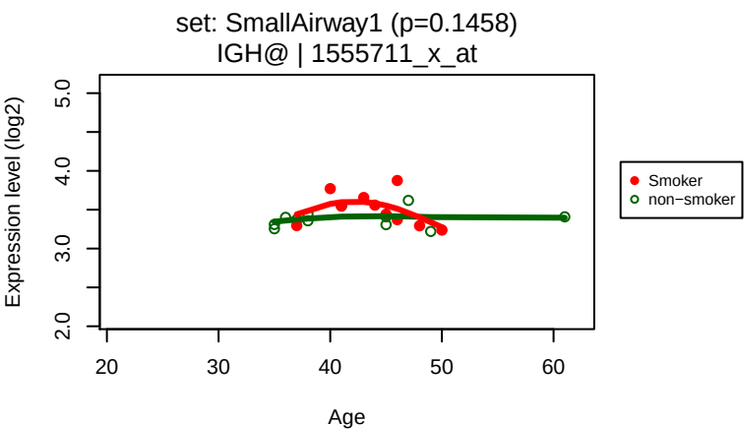
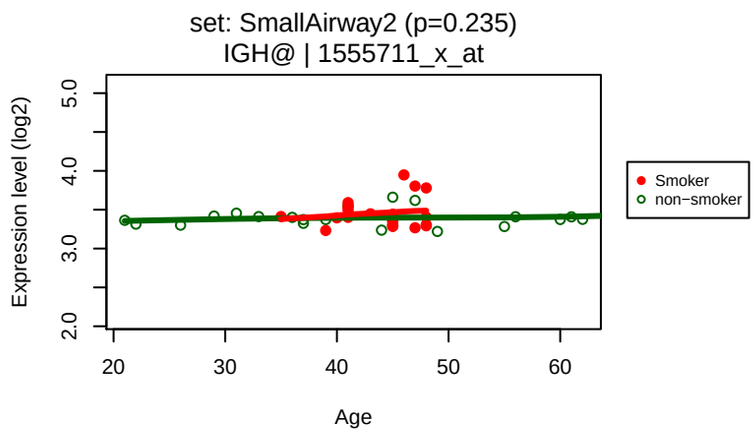
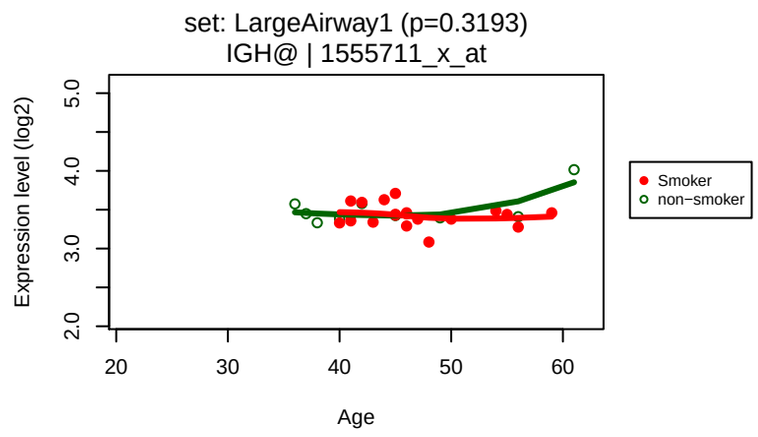

Gene expression of smokers and nonsmokers
gene: IGH@ | 1555711_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1458
p=0.235
p=0.4831
p=0.3193
p=0.6386
p=0.5535
p=0.4614
p=0.8213
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

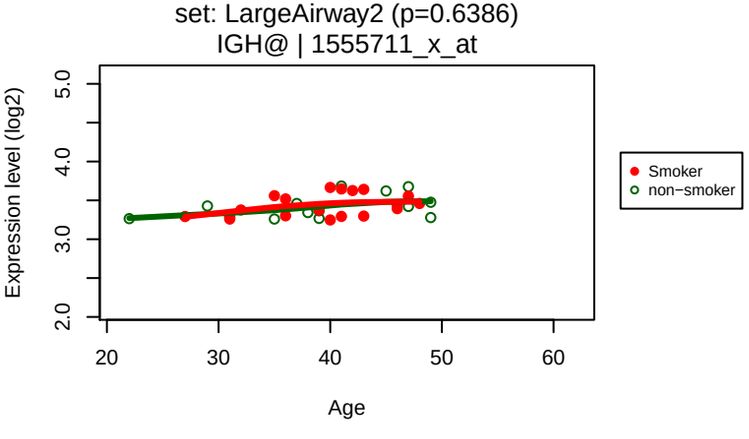

Gene expression ratio (smokers/nonsmokers)
gene: IGH@ | 1555711_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1458
p=0.235
p=0.4831
p=0.3193
p=0.6386
p=0.5535
p=0.4614
p=0.8213
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

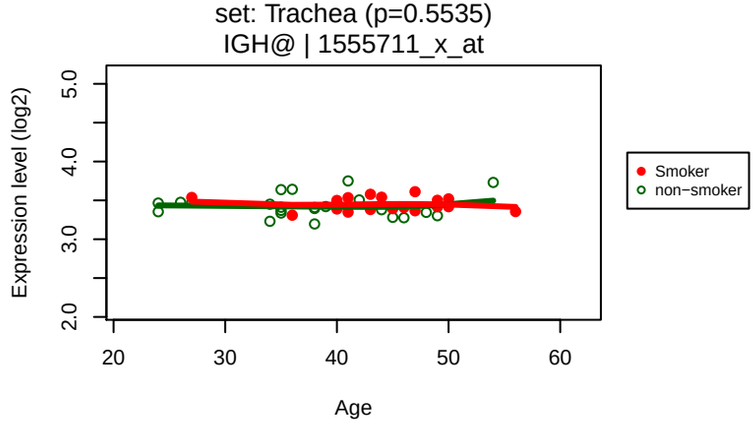

Gene expression ratio in Males
gene: IGH@ | 1555711_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.697
p=0.2544
p=0.1779
p=0.6826
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGH@ | 1555711_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3775
p=0.4857
p=0.4024
p=0.9873
SmallAirway2
LargeAirway2
Trachea
Macrophage2

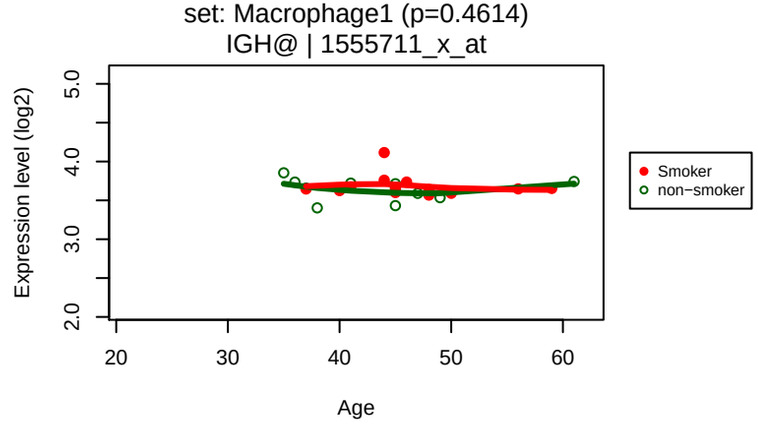

set: SmallAirway1 (p=0.1458)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker

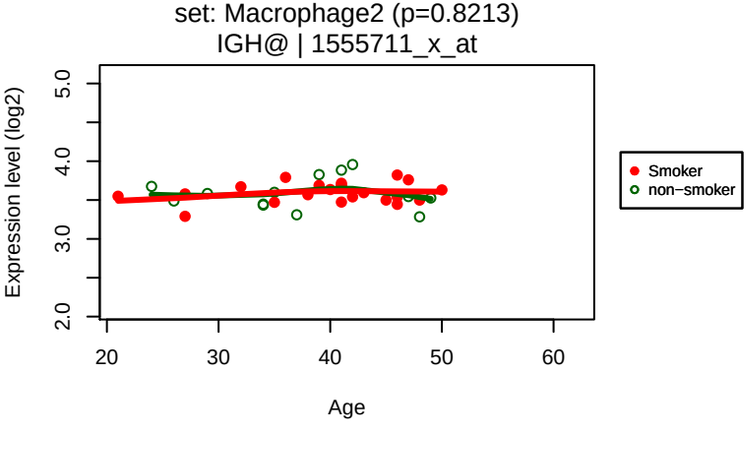

set: SmallAirway2 (p=0.235)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker

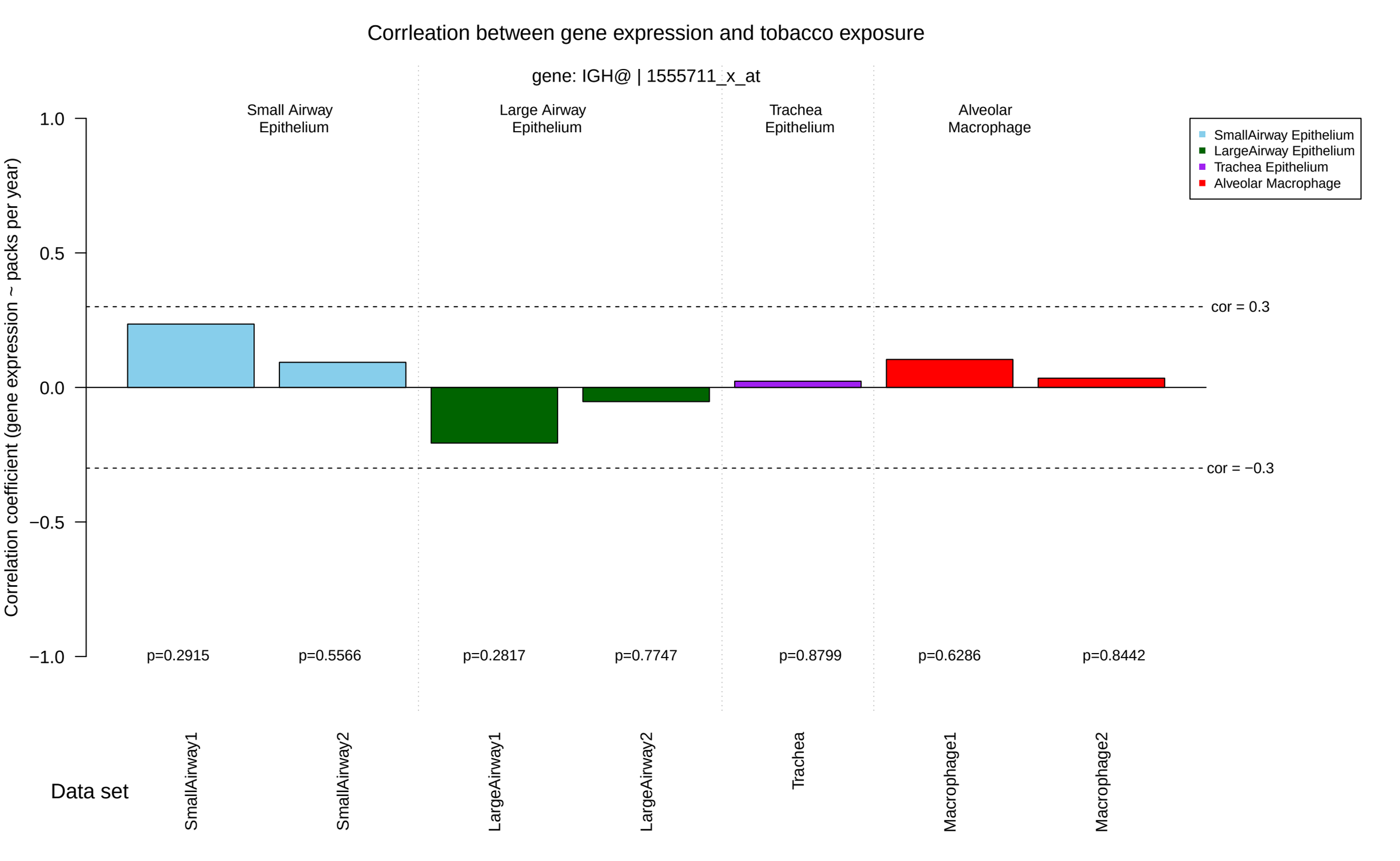

set: LargeAirway1 (p=0.3193)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6386)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5535)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4614)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8213)
IGH@ | 1555711_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGH@ | 1555711_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2915
p=0.5566
p=0.2817
p=0.7747
p=0.8799
p=0.6286
p=0.8442
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2